Sneha Kulkarni

M.Tech Machine Design · Applied ML Engineer
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Summary

- Applied machine-learning engineer with 5 years of experience in predictive maintenance for manufacturing.

Education

Western University of Engineering, Pune 2017 – 2019
M.Tech in Machine Design – CGPA: 8.7/10

Western University of Engineering, Pune 2013 – 2017
B.E. in Mechanical Engineering – CGPA: 8.2/10

Technical Skills

- **Languages:** R, MATLAB, Julia, C++
- **Machine Learning:** Machine Learning, Deep Learning, Statistics, Computer Vision
- **Data & Tools:** Tableau, Microsoft Excel, Simulink
- **Engineering:** SolidWorks, ANSYS, AutoCAD

Experience

ML Engineer – Apex Motors, Pune Jun 2019 – May 2024

- Built vibration-based failure models in MATLAB that flag 87% of gearbox faults 10 days early.
- Cut unplanned downtime on 3 assembly lines by 23% in the first year of deployment.
- Trained a defect-detection model on 18,000 images of cast parts with 94% recall.

Projects

Bearing Fault Classifier | MATLAB · Deep Learning Toolbox 2023
github.com/sneha-kulkarni-example/bearing-faults

- Classifies 4 fault types from 12 kHz vibration signals at 96% accuracy.

Tool-Wear Forecasting | R · tidymodels 2022
github.com/sneha-kulkarni-example/tool-wear

- Forecasts cutting-tool wear 50 cycles ahead with a mean error of 4%.

Achievements and Positions of Responsibility

- Best Paper – National Conference on Smart Manufacturing (2023)